Some solutions to *Introduction to Symplectic Topology*

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I. Introduction

Worked solutions to problems in McDuff and Salamon's book, *Introduction to Symplectic Topology*.

Here are the current tallies of problems solved:

- Chapter 1: X/X
- Chapter 2: X/X
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II. Chapter 1

Solution II.1 (Problem 1.1.5)**.** We are assuming that $(x, y) : I \to \mathbb{R}^{2n}$ is a trajectory such that

$$\dot{x}_j(t) = \frac{dH}{dy_j}(t, x(t), y(t)) \quad \text{and} \quad \dot{y}_j(t) = -\frac{dH}{dx_j}(t, x(t), y(t)) \tag{1}$$

We are also assuming that $\det \frac{d^2H}{dy_i dy_j} \neq 0$. Consider the equation, $v = \frac{dH}{dy}(t, x, y)$, suppose (v_0, t_0, x_0, y_0) satisfies the equation. The condition and implicit function theorem implies that in a neighbourhood of (v_0, t_0, x_0), we can write $y = G(v, t, x)$ with $y_0 = G(v_0, t_0, x_0)$. We then define

$$L(t, x, v) = \sum_j v_j G_j(v, t, x) - H(t, x, G(v, t, x)) \tag{2}$$

in a neighbourhood of (t_0, x_0, v_0). Suppose we consider the point $(v_0, t_0, x_0, y_0) = (\dot{x}(t_0), t_0, x(t_0), y(t_0))$ for some t_0. Clearly, such a point will satisfy the above criterion, so we can define L in a neighbourhood of

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$(\dot{x}(t_0), t_0, x(t_0))$. We have

$$\frac{dL}{dv_j}(t, x(t), \dot{x}(t)) = G_j(\dot{x}(t), t, x(t)) + \sum_i \left[\dot{x}_i(t)\frac{dG_i}{dv_j}(\dot{x}(t), t, x(t)) - \frac{dH}{dy_i}(t, x(t), y(t))\frac{dG_i}{dv_j}(\dot{x}(t), t, x(t))\right] \tag{3}$$

$$= G_j(\dot{x}(t), t, x(t)) = y_j(t) \tag{4}$$

which means that

$$\frac{d}{dt}\frac{dL}{dv_j}(t, x(t), \dot{x}(t)) = \dot{y}_j(t) = -\frac{dH}{dx_j}(t, x(t), y(t)) \tag{5}$$

From here, note that

$$\frac{dL}{dx_j}(t, x(t), \dot{x}(t)) = \sum_i \left[\dot{x}_i(t)\frac{dG_i}{dx_j}(\dot{x}(t), t, x(t)) - \frac{dH}{dy_i}(t, x(t), y(t))\frac{dG_i}{dx_j}(\dot{x}(t), t, x(t))\right] - \frac{dH}{dx_j}(t, x(t), y(t)) \tag{6}$$

$$= -\frac{dH}{dx_j}(t, x(t), y(t)) \tag{7}$$

which implies that the prior equation is exactly the Euler-Lagrange equation, as desired.

Solution II.2 (Problem 1.1.20)**.** Note that

$$\omega_0(X_F, X_G) = dF(X_G) = X_G(F) = -(\nabla F)^T J_0 \nabla G = \{F, G\} \tag{8}$$

Thus, if $X = X_F$ and $Y = X_G$ are symplectic vector fields, then

$$X_{\omega_0(X,Y)} = X_{\{F,G\}} = [X_F, X_G] = [X, Y] \tag{9}$$

Solution II.3 (Problem 1.1.24)**.** In the particular case of geodesic flow, let us begin by noting that

$$\frac{d^2L(x,v)}{dv_i dv_j} = \frac{d^2}{dv_i dv_j}\frac{1}{2}\sum_{ij} g_{ij}(x)v_i v_j = \frac{1}{2}(g_{ij}(x) + g_{ji}(x)) = g_{ij}(x) \tag{10}$$

which means that the determinant is non-zero, as the metric is a non-degenerate symmetric two-tensor. Thus, we may apply the Legendre transform everywhere to get $y_k = \frac{dL}{dv_k} = \sum_j g_{kj}(x)v_j$. This means that $v_j = \sum_k g^{jk}(x)y_k$, and we get as our Hamiltonian

$$H(x,y) = \sum_j y_j v_j - L = \sum_j g^{jk}(x)y_j y_k - \frac{1}{2}\sum_{ijk\ell} g_{ij}(x)g^{ik}(x)g^{j\ell}(x)y_k y_\ell \tag{11}$$

$$= \sum_{jk} g^{jk}(x)y_j y_k - \frac{1}{2}\sum_{ik} g^{ik}(x)y_k y_i \tag{12}$$

$$= \frac{1}{2}\sum_{jk} g^{jk}(x)y_j y_k = \frac{1}{2}\langle y, g^{-1}(x)y\rangle \tag{13}$$

which implies that Hamilton's equations become

$$\dot{y}_k = -\frac{1}{2}\sum_{ij}\frac{dg^{ij}(x)}{dx_k}y_i y_j \quad \text{and} \quad \dot{x}_k = \frac{1}{2}\sum_j g^{jk}(x)y_j \tag{14}$$

To prove that $|\cdot|_g$ is constant along geodesics, we can show that the smooth function $\frac{1}{2}|v|_g^2 = \frac{1}{2}\langle v, g(x)v\rangle$ is constant along geodesics, $(x,v) = (x(t), \dot{x}(t))$. But of course, $v = g^{-1}(x)y$, so if $(x(t), y(t))$ is the phase space trajectory, then $\dot{x}(t) = g^{-1}(x(t))y(t)$, and

$$\frac{1}{2}|\dot{x}(t)|_g^2 = \langle g^{-1}(x(t))y(t), y(t)\rangle = H(x(t), y(t)) \tag{15}$$

We know that the Hamiltonian H will remain constant over geodesics, so the norm velocity will as well.

Solution II.4 (Problem 1.2.2)**.** Note that for Hamiltonian functions H and H' such that $S = H^{-1}(c) = (H')^{-1}(c')$ for regular values c and c', the Hamiltonian vector fields $X_H(z)$ and $X_{H'}(z)$ lie in the rank-one subbundle of T_zS given by J_0N_zS, where N_zS is the normal bundle at z. Note that $\omega(X_H(z), J_0X_H(z)) = -||X_H(z)||^2$, so we can define $\lambda(z) = -\omega(X_{H'}(z), J_0X_H(z))||X_H(z)||^{-2}$, provided that $X_H(z) \neq 0$, which we may assume as c is a regular value, implying H has non-zero derivative at every $z \in S$. We then have $X_{H'}(z) = \lambda(z)X_H(z)$. Note that λ is always non-zero. Consider the function $\lambda \circ z : \mathbb{R} \to \mathbb{R}$, where z is the solution of the H -system. By existence of uniqueness, the ODE $\dot{\tau}(t) = (\lambda \circ z)(\tau(t))$ has a unique solution with $\tau(0) = 0$ on the same time interval that z is defined on (all of $\mathbb{R}$). It is a reparametrization as $\dot{\tau}(t) \neq 0$, as $\lambda \circ z$ never is 0. To find T' such that $\tau(kT') = kT$ for all $k \in \mathbb{Z}$, note that we can of course find unique T' such that $\tau(T') = T$. Let $\sigma(t+kT') = \tau(t)+kT$ for $t \in [0, T']$ and $k \in \mathbb{Z}$. Note that $\sigma(kT') = \tau(0)+kT = kT$ while $\sigma(T'+(k-1)T') = \tau(T')+(k-1)T = kT$, so this function is well-defined and continuous. It is smooth on each interval $(kT', (k+1)T')$. In addition,

$$(\lambda \circ z)(\sigma(t + kT')) = \lambda(z(\tau(t) + kT)) = \lambda(z(\tau(t))) = \dot{\tau}(t) = \dot{\sigma}(t + kT') \tag{16}$$

on each of these open intervals. Hence, by uniqueness, $\sigma = \tau$ on each of the open intervals, and by continuity, they will be equal everywhere. In particular, we have $\sigma(kT') = kT$ for all k, as desired. It follows immediately that

$$\frac{d}{dt}(z \circ \tau)(t) = \dot{z}(\tau(t))\dot{\tau}(t) = \lambda(z(\tau(t)))X_H(z(\tau(t))) = X_{H'}(z(\tau(t))) \tag{17}$$

so that $z \circ \tau$ is a time-periodic solution of the H' -system, as desired.

III. Chapter 2

Solution III.1 (Problem 2.1.2)**.** The symplectic form $(-\omega) \oplus \omega$ on vector space $V \times V$ acts like

$$(-\omega \oplus \omega)((v_1, v_2), (w_1, w_2)) = \omega(v_2, w_2) - \omega(v_1, w_1) \tag{18}$$

for $(v_1, v_2), (w_1, w_2) \in V \times V$. It follows immediately that it is a skew-bilinear form, and is non-degenerate because if

$$(-\omega \oplus \omega)((v_1, v_2), \cdot) = \omega(v_2, \cdot) - \omega(v_1, \cdot) = 0 \tag{19}$$

then plugging in $(w_1, w_2) = (w, 0), (0, w)$ implies $v_1 = v_2 = 0$, from non-degeneracy of ω. Note

$$(-\omega \oplus \omega)((w_1, w_2), (v, \Psi v)) = \omega(w_2, \Psi v) - \omega(w_1, v) \tag{20}$$

Thus, if Γ_Ψ is Lagrangian, then for any $(w, \Psi w)$, we will have for all v, $\omega(\Psi w, \Psi v) = \omega(w, v)$, implying Ψ is a symplectomorphism. Conversely, if Ψ is a symplectomorphism, then we will clearly have $\Gamma_\Psi \subset \Gamma_\Psi^\omega$ from what we just stated, for Γ_Ψ is isotropic, and moreover, $\dim \Gamma_\Psi = \dim V$, so from Lemma 2.1.1, it is Lagrangian.

Solution III.2 (Problem 2.1.10)**.** Note that for $w \in W$ and $v \in W \cap W^\omega$, $\omega(w, v) = 0$, so $\overline{\omega}([w_1], [w_2]) = \omega(w_1, w_2)$ is well-defined, clearly this is skew-bilinear, and to see that it is non-degenerate, note that if $\overline{\omega}([w], [v]) = \omega(w, v) = 0$ for all $[v]$, then $w = 0$, so $[w] = 0$.

Solution III.3 (Problem 2.1.11)**.** Note that the complex matrix iA is self-adjoint, and therefore is diagonalizable with real eigenvalues. Let $w_1, \ldots, w_{2n}$ be an orthonormal eigenbasis. In particular, we have $iAw_j = \lambda_j w_j$ for each j, so $Aw_j = -i\lambda_j w_j$: A has purely imaginary eigenvalues. Taking complex conjugates, using the fact that A is a real matrix, we have $A\overline{w_j} = i\lambda_j \overline{w_j}$, so if $w_j = v_j + iu_j$, then $v_j - iu_j$ is also an eigenvector. Note that $\lambda_j \neq 0$ for all j, so our eigenvectors will be exactly $v_j \pm iu_j$ for $j = 1, \ldots, n$ with eigenvalues $\mp i\lambda_j$ respectively (assume WLOG $\lambda_j > 0$). Our claim is that $v_1, \ldots, v_n, u_1, \ldots, u_n$ forms a symplectic basis.

Recall that eigenvectors of a self-adjoint matrix corresponding to different eigenvalues will be orthogonal. Indeed, if B is some such matrix, v is a λ eigenvalue and w is a λ' eigenvalue, then

$$(\lambda - \lambda')\langle v, w\rangle = \langle Av, w\rangle - \langle v, Aw\rangle = 0 \tag{21}$$

Therefore, the $v_j \pm iu_j$ are all pairwise orthogonal. Thus, for $i \neq j$,

$$0 = \langle v_i + iu_i, v_j \pm iu_j\rangle_{\mathbb{C}} = (\langle v_i, v_j\rangle \pm \langle u_i, u_j\rangle) + i(\langle u_i, v_j\rangle \pm \langle v_i, u_j\rangle) \tag{22}$$

which means that

$$\langle u_i, u_j\rangle = \langle v_i, v_j\rangle = \langle u_i, v_j\rangle = \langle v_i, u_j\rangle = 0 \tag{23}$$

In addition,

$$0 = \langle v_i + iu_i, v_i - iu_i\rangle_{\mathbb{C}} = (\langle v_i, v_i\rangle - \langle u_i, u_i\rangle) + 2i\langle u_i, v_i\rangle \tag{24}$$

so that $\langle u_i, v_i\rangle = 0$ and $\langle v_i, v_i\rangle = \langle u_i, u_i\rangle$. Because $A(v_i + iu_i) = -i\lambda_i(v_i + iu_i)$, with A, u_i, v_i, λ_i real, we have $Av_i = \lambda_i u_i$ and $Au_i = -\lambda_i v_i$. Thus,

$$\omega(v_i, u_j) = \langle Av_i, u_j\rangle = \lambda_i\langle u_i, u_j\rangle = \lambda_i\langle u_i, u_i\rangle\delta_{ij} \tag{25}$$

and

$$\omega(u_i, u_j) = \langle Au_i, u_j\rangle = -\langle v_i, u_j\rangle = 0 = \omega(v_i, v_j) \tag{26}$$

so if we normalize to ensure that $\langle v_i, v_i\rangle = \langle u_i, u_i\rangle = \lambda_i^{-1}$, then we will have a symplectic basis.

Solution III.4 (Problem 2.1.12)**.** Given such a pair (W, β), pick some basis $w_1, \ldots, w_{2n+p}$ and let $A_{ij} = \omega(w_i, w_j)$. Clearly, A is a skew-symmetric matrix. Let $V = \mathrm{Ker}(A)$. Pick some $v \in V^\perp$ and note that for $w \in \mathrm{Ker}(A)$, we have $\langle Av, w\rangle = -\langle v, Aw\rangle = 0$, so $Av \in \mathrm{Ker}(A)^\perp$. Thus, we have restricted operator $A|_{V^\perp} : V^\perp \to V^\perp$. Note that this matrix is injective, hence invertible, so the form $\omega(v, w) = \langle Av, w\rangle$ defined on the subspace $V^\perp$ makes it a symplectic vector space of dimension $2n$. Let $e_1, \ldots, e_n, f_1, \ldots, f_n$ be a symplectic basis. We can then choose a basis for V of the form $g_1, \ldots, g_p$. The corresponding basis for W will be one with the desired properties.

Solution III.5 (Problem 2.1.13)**.** We go through each case:

- In the case that W is isotropic, so $W \subset W^\omega$, suppose $w_1, \ldots, w_p$ is a standard basis. Let Λ be a Lagrangian subspace containing W. We can extend $w_1, \ldots, w_p$ arbitrarily to a basis $w'_1, \ldots, w'_{p'}$ for Λ. Note that because $\Lambda = \Lambda^\omega$, $\omega(w'_i, w'_j)$ vanishes for all pairs of vectors in the basis, so the basis is still standard. From here, we may invoke Lemma 2.1.5 to extended this basis to a symplectic basis for all of V.

- In the case that W is coisotropic, let $v_1, \ldots, v_n, u_1, \ldots, u_n, w_1, \ldots, w_p$ be a standard basis. It is easy to see that $w_1, \ldots, w_p \in W^\omega$ forms a basis for this subspace. Clearly, U spanned by the v_i and u_j is symplectic, so $V = U \oplus U^\omega$ with $W^\omega \subset U^\omega$. Note that U^ω is symplectic, and clearly, as a subspace of U^ω, W^ω is Lagrangian, so we can extend to a basis for U^ω. Combining together these bases of U and U^ω yields an symplectic basis for V which is an extension of the original basis.

- Finally, in the case that W is symplectic, then W^ω is also symplectic, so we can just find any symplectic basis for W^ω, and combining the bases will yield the desired symplectic basis for V.

Solution III.6 (Problem 2.1.14)**.** Note that W must contain some w such that $\omega(w, x) = 0$ for all $x \in W$, otherwise $W \cap W^\omega = 0$ so that W is symplectic, hence even-dimensional, which is a contradiction. On the other hand, $\dim(V) = \dim(W) + \dim(W^\omega)$, so $\dim(W^\omega) = 1$, which means that $W^\omega \subset W$, as desired.

IV. Chapter 3